

WEEK 4 – SPL 4 – POLYMERS

FOR STATISTICS ABOUT POLYMERS CHECK THE PPT !!

THERMOPLASTICS – Heat up more – becomes more plastic – more easy to deform – squeeze to conform new shape – cool it down = new product :)

THERMOSETTING – low viscosity, HIGH TEMP , THERMAL CAPABILITIES THAN THERMOPLASTICS!, can NEVER BE REMELTED!! This is an issue for car tyres as we cant heat up and make into a new material and recycle..

Property / Characteristic	Thermoplastics	Thermosets
Molecular Structure	Mostly linear molecules, with a degree of branching and side chains	Three-dimensional cross-linked network
Recyclability	Re-meltable and recyclable	Cannot be remelted once cured (non-recyclable)
Processing Behaviour	Softens when heated, solidifies when cooled	Starts as a liquid, undergoes chemical curing to harden
Examples	Polyethylene, Polypropylene, ABS, PET, Nylon	Epoxy resin, Unsaturated Polyester
Thermal Properties	Poor high-temp resistance	Good high-temp resistance for polymers
Initial Form	Usually supplied as pellets	Supplied as liquid or pre-polymer resin
Flow During Processing	Flows repeatedly when reheated	Flows once, then cures irreversibly
Viscosity During Processing	High viscosity (like treacle)	Low viscosity (like water) before curing
Typical Use Cases	Packaging, automotive panels, toys	Electrical insulation, adhesives, structural composites

1. What does the “Scale of the Industry” data tell us about the economic and employment impact of the polymer sector in Europe, and how might this influence engineering design choices?

The "Scale of the Industry" slide illustrates the **massive economic and employment footprint** of the polymer sector in Europe (EU, 2012 data):

- Over **1.5 million people** were employed in polymer-related industries.
- The value of **polymer plastic production €87 billion**.
- **Conversion** of these plastics into usable parts and products reached a **€201 billion**.

Polymers:

- **Mass production** due to low cost and scalability
- **Lightweight solutions** for transportation and energy efficiency
- **Customisable properties** to meet specific design requirements

Shape Change Process in Polymers

- Polymers are usually **processed using a moulding** operation involving shape changes in a liquid state.
- At **much lower temperatures than those required for metals**. Polymers melt and flow at much **lower temperatures** than metals, **reducing energy consumption and thermal stress on tooling**.
- At lower pressures and hence lower tooling costs when compared to metals.

Moulding of Thermoplastics and Thermosets

- **THERMOPLASTIC** – heat polymer up , squeeze (mold) it into a chamber under pressure (still quite viscous)
Supplied as solid pellets
- **THERMOSET** – opp of thermoplastic, starts as a really low viscosity fluid, pump it into a chamber, heat up chamber – comes out as solid e.g., epoxy resin more viscous than water but less than e.g., treacle or honey
Supplied as a liquid or semi-liquid resin

Moulding of Thermoplastics and Thermosets



- The difference in processing between thermoplastics and thermosets is that:
 - Thermosets solidifies by virtue of a chemical reaction where the liquid transforms to a solid by a process called curing - often requiring heat or a hot mould.
 - Thermoplastics solidify by cooling in the mould. Thermoplastics therefore generally require cold (or cool) moulds.
- At processing temperatures, the viscosity of thermosets is often very low (like water) while that of a thermoplastics is much higher (like treacle).

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INJECTION MOULDING

Injection moulding is a **fast, automated process used to convert granular thermoplastic materials into complex, detailed parts**. It's widely used in mass manufacturing e.g.,

Step-by-Step Process:

1. Feeding:

- Plastic granules (usually thermoplastics) are loaded into a hopper.
- They drop into a barrel with a rotating screw.

2. Heating & Melting:

- As the screw rotates, it conveys the granules forward.
- The friction from the screw + external heaters causes the granules to melt into a viscous fluid.

3. Injection:

- Once a full shot of molten polymer is ready, the screw acts as a plunger, pushing the melt through a nozzle into the mould cavity.

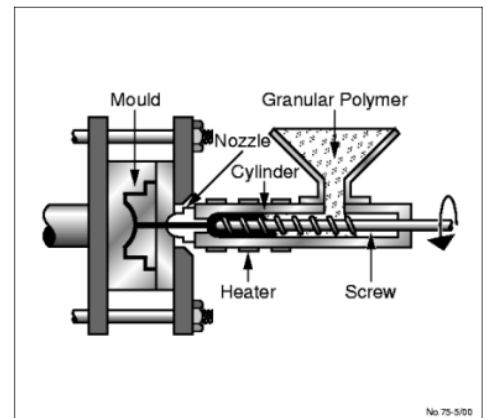
4. Cooling & Ejection:

- The polymer cools and solidifies in the mould (which is kept cold).
- The mould then opens, and the part is ejected.

Injection Moulding



- This process is designed to facilitate high speed processing of plastics into parts (which may be intricate).
- A feed unit introduces the polymer into the barrel.
- This barrel contains a screw that transports the granules into a chamber.
- The screw does work on the polymer which heats it up and induces melting - a process assisted by additional heating as necessary.



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Other forms of Injection Moulding

Gas-Assisted Injection Moulding

- Involves injecting gas (usually nitrogen) into the mould after the polymer.
- The gas pushes the molten plastic outward, creating hollow sections inside the part.
- **Advantages:**
 - Reduces material usage
 - Makes parts lighter
 - Helps produce thicker, more uniform walls
- Common in: back of TV

Co-injection moulding - mould multiple polymers into a single part

Advantages:

- Combines cost-efficiency and performance
- Useful for products needing aesthetic finish + strength

Scrim

Uses **multiple injection barrels** with **one tool**.

After filling the mould, **oscillating pressure** is applied **before the polymer solidifies**.

Advantage:

- Create **preferred molecular alignment** (orientation)
- **Reduce voids** and improve part quality

Downside: It's **expensive** and can **damage moulds** due to pressure cycling.

External Gas Injection Moulding: THERMOPLASTIC

- A **thin layer of gas** is injected **between the outer surface of the molten polymer and the mould wall**.
- The gas pushes the polymer **tightly against the opposite mould surface**.
- This helps to **replicate fine surface details** while maintaining **hollow interiors**.

Used for:

- Parts where **external appearance and surface detail** are important.
- Creating **uniform outer surfaces** with **reduced material use**.
-

Internal Gas Injection Moulding:

- The gas is injected **into the core of the molten plastic**, while it's still inside the mould.
- The gas **pushes the plastic outward**, forcing it to fill the mould extremities.
- It also helps create **hollow sections**, reducing weight and material.

Used for:

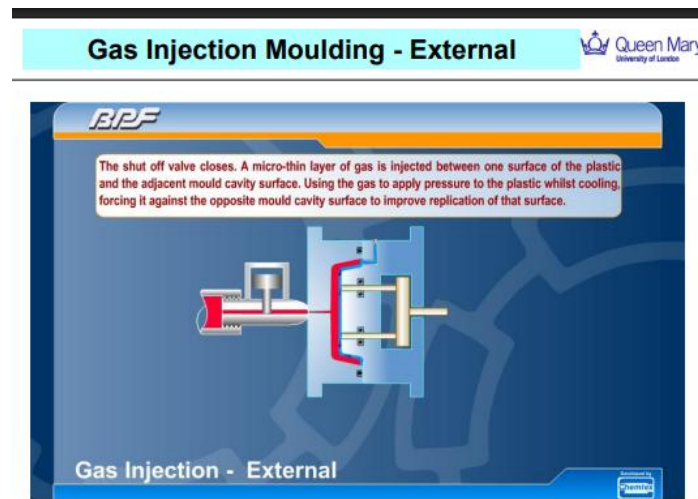
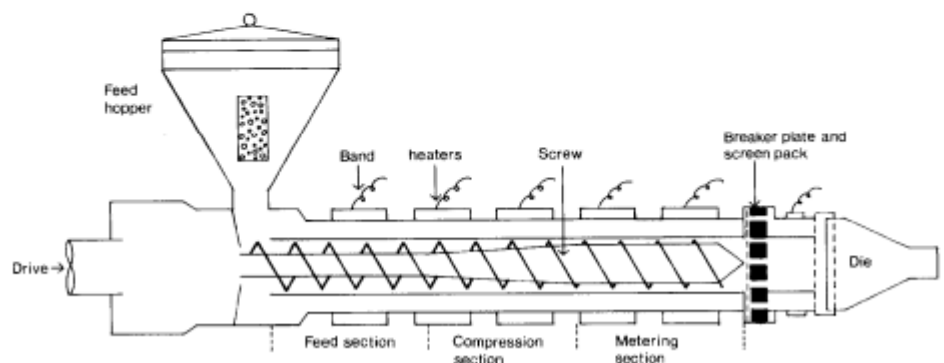
- Producing **hollow parts** with **thick sections** (e.g., handles, tubes).
- Ensuring complete fill of complex geometries without sink marks.

EXTRUSION - **continuous polymer shaping process** used to produce parts with a **constant cross-section**, such as **pipes, tubes, window frames, and plastic sheets**.

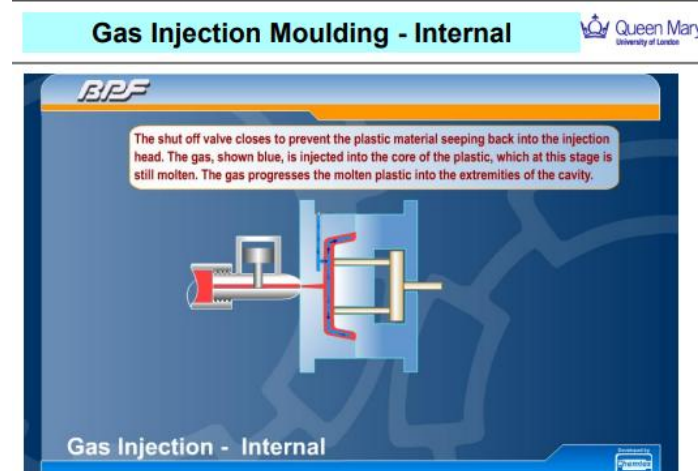
EXTRUSION PROCESSES - thermoplastic

Extrusion Processes

- This is similar to the metal forming process and involves forcing a molten polymer through a die, chilling rapidly on exit to preserve the cross section and allowing the process to continue indefinitely.



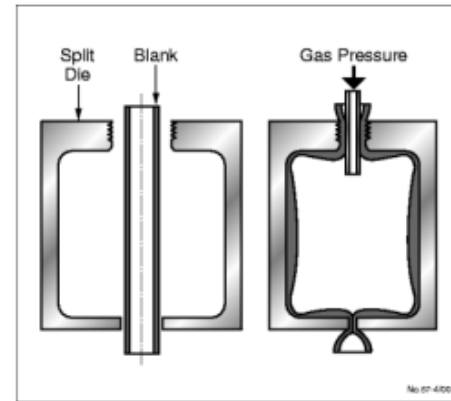
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Blow Moulding - e.g. bottles

1. A **molten polymer tube** (called a **parison**) is extruded vertically.
2. The **parison is enclosed** within a **split mould**.
3. **Compressed air or gas** is then blown into the parison.
4. The pressure causes the polymer to **expand and press against the mould walls**, taking its shape.
5. The plastic cools and **hardens** in that form.
6. The mould opens and the final part is **ejected**

- Blow moulding was originally developed for glasses and adapted successfully to polymers.
- The material (the parison) is supplied hot in tubular form enclosed by the split mould, and forced to adopt the shape by air (or gas) pressure. The process is ideal for producing simple shapes, like containers where the surface finish is only critical on the outside.
- The process is cheap as tooling costs are low as the pressures are not high and only small clamping forces required.



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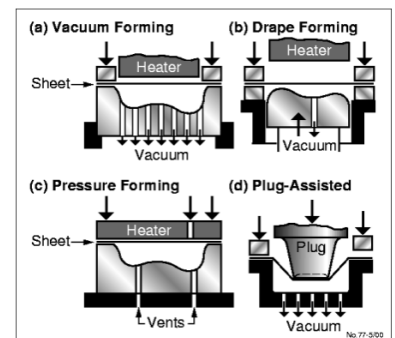
- A **vacuum is applied beneath the mould**, sucking the sheet into its shape.
 - The **sheet cools and hardens, taking on the mould's contours**.
- ✓ (a) **Vacuum Forming** – Basic form using suction only
 - ✓ (b) **Drape Forming** – Sheet sags over the mould before vacuum is applied
 - ✓ (c) **Pressure Forming** – Uses vacuum plus extra air pressure on top
 - ✓ (d) **Plug-Assisted Forming** – A plug pushes the sheet into the mould before vacuum is applied, improving accuracy

Vacuum Forming

Vacuum Forming

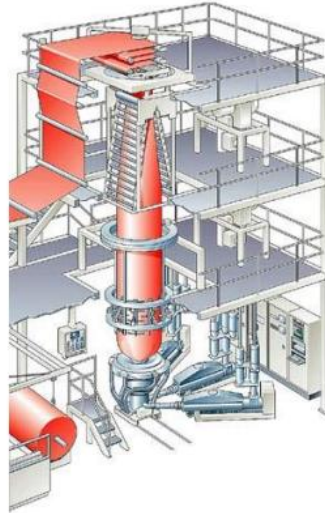


- Here the required shape is formed using air pressure. A polymer sheet is drawn as a hot formable sheet onto a tool via the application of a vacuum.
- Again the process is cheap with a low cost tool due to the low pressures. The material requirements for the polymer will be similar to that required for blow moulding.
- The complexity of shape that can be made in this way is obviously limited but large shapes can be made economically.



BLOWN FILM – PLASTIC BAG

- An extruder is fitted with an annular die, pointing upwards.
- This tube is blown up with air, and is drawn up in continuous process. The air is contained in the bubble by collapsing rolls at the top.
- Like blow moulding this induces a bi-axial orientation, which is then made permanent by the crystallisation process which freezes the orientation in place.
- This process can form either sheets or bags. Bags need to have the edges cut off and require rewinding into two different rolls. The bags require heat sealing at regular intervals and regular perforations to permit tear off or cutting.



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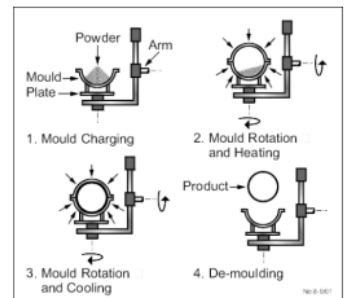
Rotational Moulding

- A pre-measured amount of thermoplastic powder is placed inside a hollow metal mould.
- The mould is sealed.
- As the mould rotates, the plastic powder melts and coats the interior surface evenly.
- Powder - melted and distributed, the mould is cooled, typically by air or water mist.
- The mould is then opened, and the solidified part is removed.

e.g., water tanks, storage bins and etc.

Rotational Moulding

- Some simple shapes may be produced by rotating a large heated mould with a charge of polymer.
- The polymer inside the mould melts and coats the inside of the mould.
- This will then be allowed to cool whilst still rotating allowing a uniform build up of polymer and ultimately a part which has conformed to the mould tool.



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Cost Comparison

INJECTION MOULDING – MACHINERY EXPENSIVE BUT EASY TO PRODUCE A BATCH

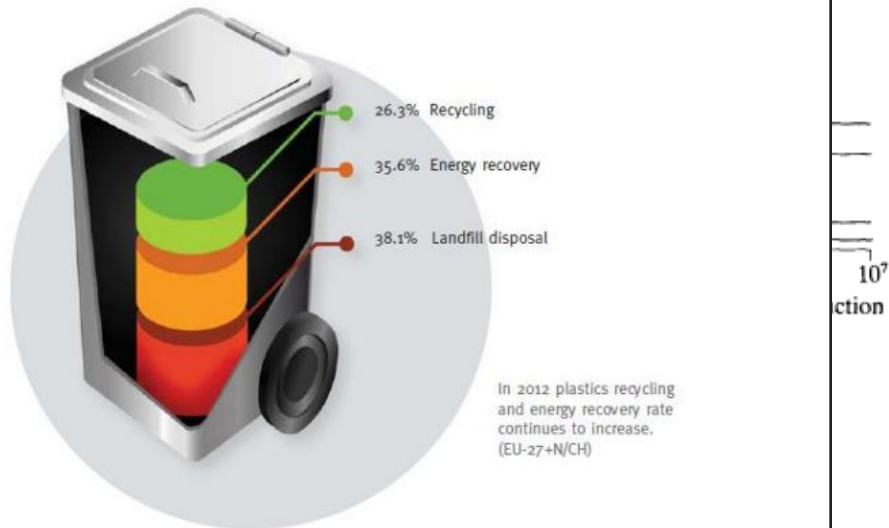
Cost Comparison for Different Processes

- In general terms the capital costs associated with injection moulding (machines and tools) are higher than blow moulding which in turn are higher than vacuum forming.
- As these costs have to be recovered from the production of parts, the cost burden per part for injection moulding is very high if the production volume is low.
- However, the cost of producing each part if the capital costs are ignored is much less for injection moulding than other processes due to a rapid production and process automation.

Cost Comparison

- *This leads to part/cost versus production volume plots of the form shown in the figure. However process volumes are seldom known in advance beyond very vague limits.*

2012 Polymer Recycling in the EU+2



✓ Thermoplastics vs. Thermosets — Process Compatibility Table

Process	Thermoplastics	Thermosets	Notes
Injection Moulding	✓ Yes	✗ No	High-speed, melt–cool–repeat cycle works for thermoplastics only
Gas-Assisted Injection	✓ Yes	✗ No	Advanced injection moulding — requires remelting
Co-Injection / Scrim	✓ Yes	✗ No	Based on standard thermoplastic injection
Extrusion	✓ Yes	✗ No	Continuous process for pipes, sheets, film — only works with meltable materials
Blown Film	✓ Yes	✗ No	Based on extrusion — creates films, bags
Vacuum Forming	✓ Yes	✗ No	Uses softened sheets of thermoplastics
Blow Moulding	✓ Yes	✗ No	Hollow parts from tubes of melted plastic
Rotational Moulding	✓ Yes	✗ Rarely	Works with powdered thermoplastics (e.g., polyethylene)
Compression Moulding	✓ Sometimes	✓ Yes	Best suited to thermosets (epoxy, polyester) — but can mould thermoplastics in some cases
Transfer Moulding	✗ No	✓ Yes	Exclusively used for thermosets (advanced electronics, encapsulation)

Method	Advantages	Disadvantages
Injection Moulding	High precision and repeatability Fast cycle times for mass production Good surface finish	High tooling cost Limited to thermoplastics Difficult for very large parts
Gas-Assisted Injection Moulding	Hollow parts with less material Reduces warping and sink marks Good for complex shapes	More complex tooling Process control is critical Only for thermoplastics
Extrusion	Continuous process Excellent for long, uniform profiles Low material waste	Limited to simple cross-sections Surface finish may need post-processing

Blown Film Extrusion	Produces thin, biaxially oriented films High throughput Ideal for packaging	Limited to thin film products Requires precise control
Blow Moulding	Great for hollow containers (e.g., bottles) Low cost for large volumes Good for thin-walled parts	Poor control over wall thickness Limited to simple shapes
Vacuum Forming	Low tooling costs Fast setup for low-volume production Simple equipment	Limited to shallow parts Poor detail resolution Only one good surface
Rotational Moulding	Uniform wall thickness Ideal for large hollow items No seams or welds	Long cycle time Lower detail accuracy Not suitable for small intricate parts
Compression Moulding	Good for thermosets Low scrap rate Can mould large parts	Limited to simple geometries Slower than injection moulding Labour intensive

What processing route is usually used to make a plastic shampoo bottle?

Select one:

- ☐ a. Injection moulding
- ☐ b. Extrusion blow moulding
- ☐ c. Rotational moulding
- ☒ d. Injection blow moulding ✖

Your answer is incorrect.

The correct answer is: Extrusion blow moulding

What processing route is usually used to make a 2 litre carbonated drinks bottle?

Select one:

- ☐ a. Rotational moulding
- ☒ b. Extrusion blow moulding ✖
- ☐ c. Injection blow moulding
- ☐ d. Injection moulding

Your answer is incorrect.

The correct answer is: Injection blow moulding

Feature/Aspect	Injection Blow Moulding	Extrusion Blow Moulding
Used for	High-precision containers (e.g., carbonated drink bottles, pharma bottles)	Larger or simpler hollow products (e.g., shampoo bottles, detergent containers)
Preform shape	Starts with an <i>injection-moulded preform</i> (precise neck & threads)	Starts with a <i>tube (parison)</i> extruded from a die
Neck finish	Excellent dimensional control – smooth, threaded, leak-proof	Less accurate – rougher or simpler necks
Wall thickness control	Very uniform wall thickness	Variable wall thickness unless controlled with die design
Material	Often PET (polyethylene terephthalate)	Often HDPE, LDPE, or PP
Cost & speed	Higher precision, lower speed, more expensive	Faster, cheaper, great for mass production
Examples	Water/soda bottles, small pharma vials	Shampoo bottles, milk jugs, fuel tanks

What processing route is usually used to make the plastic cap of a plastic drinking bottle?

Select one:

- ☐ a. Rotational moulding
- ☐ b. Injection blow moulding
- ☒ c. Injection moulding ✓
- ☐ d. Extrusion blow moulding

Which process involves combining a resin, reinforcement, catalyst and additives in a premixed optimum proportion; the materials are then cut into a shape that is similar to the mould dimensions and the mould is then closed, pressurised and heated to form and cure the part?

Select one:

- ☒ a. Sheet moulding compound ✓
- ☐ b. Resin transfer moulding
- ☐ c. Hand lay up
- ☐ d. Autoclave moulding

SPL 4 – COMPOSITES

Specific stiffness and **specific strength** are **material efficiency metrics** that compare a material's mechanical performance relative to its weight.

- **Specific Stiffness** = Young's Modulus / Density
→ Measures how stiff a material is **per unit weight**
- **Specific Strength** = Strength / Density
→ Measures how strong a material is **per unit weight**

These properties are **critical** in industries like **aerospace, automotive, and sports equipment**, where materials must be **lightweight** but still **mechanically robust**.

1. Why are composites increasingly used in aircraft like the Boeing 787, and what benefits do they provide over traditional materials?

The **aerospace industry** has shifted heavily toward composites due to their **exceptional specific strength and stiffness**, as well as their resistance to environmental degradation.

In the **Boeing 787 Dreamliner**:

- Over **50% of the structural weight** comes from **composites**, primarily **carbon fibre-reinforced polymers (CFRPs)**.

Key Benefits of Composites in Aircraft:

1. **Weight Reduction:**
 - Composites are significantly **lighter than metals**, reducing overall aircraft weight.
 - Less weight = **better fuel efficiency** and **lower emissions**.
2. **High Strength & Stiffness:**
 - CFRPs provide **excellent load-bearing capabilities** with low density.
 - Critical for wings, fuselage panels, and control surfaces.
3. **Corrosion & Fatigue Resistance:**
 - Unlike metals, composites **do not corrode** and are less prone to **fatigue failure**.
 - This **reduces maintenance costs** and extends service life.
4. **Manufacturing Flexibility:**
 - Composites can be **moulded into complex, integrated shapes**, reducing the number of rivets, joints, and fasteners.
 - Improves **aerodynamic efficiency** and reduces assembly time.

The United States plays a **leading role** in the global composites market in both **value and volume**. This highlights the country's strong adoption of composite technologies across sectors.

- The **USA accounts for 36%** of the **global composites market by value** (~€22 billion).
- It also contributes **35% of the total volume** (2.7 million metric tons).
- Composites make up **16% of the USA's structural materials market**, compared to:
 - **Steel: 76%**
 - **Aluminium: 8%**

This shows that while **metals still dominate**, composites are **steadily gaining market share** — especially in sectors requiring **high performance-to-weight ratios**.

Global Comparison – Per Capita Use:

- USA: **8–9 kg per person**
- China: **1–2 kg per person**

2. What's the difference between isotropic and anisotropy?

Isotropic Materials:

- **Same properties in all directions.**
- Includes most **metals, plastics, and ceramics**.
- Design and analysis are simpler because behaviour is **uniform** no matter how you load the material.

Anisotropic Materials (like composites):

- Properties **vary with direction**, especially with **fibre-reinforced composites**.
- The **mechanical performance depends on how the fibres are aligned**.
- The **processing route (e.g., layup orientation)** and the **final part properties** are **interdependent** — meaning you can't separate design from manufacturing.

Initial Design Considerations:

When designing with composites, ask:

- What's the **performance objective**?
- What's the **most cost-effective way** to arrange fibres and embed them in the matrix?
- What **processing method** (e.g., hand layup, filament winding) ensures correct orientation?

3. What are the three essential components of a composite material, and how does each contribute to the final performance of the part?

1. Fibre (Reinforcement):

- **The load-bearing element of the composite**
- **Provides strength, stiffness, and directional performance**
- **Common fibres:**
 - **Glass fibre** – affordable, good strength
 - **Aramid (Kevlar)** – high toughness, impact resistance
 - **Carbon fibre** – high stiffness and low density

2. Matrix (Binding Phase):

- **The glue that binds fibres together and transfers stress between them**
- **Protects fibres from mechanical damage and environmental degradation**
- **Types of matrices:**
 - **Thermosets: Epoxy, polyester** → rigid, high-temperature resistance

- Thermoplastics: PEEK, polyimide → more ductile, recyclable

3. Interface (Interphase Region): CHATGPT

- The boundary layer between fibre and matrix
- Critical for load transfer — poor bonding = weak composite
- Influenced by:
 - Surface treatment of fibres (e.g., sizing agents)
 - Manufacturing conditions (e.g., temperature, pressure, cure cycle)

JAMES: interface = weak therefore need stress transform from fibre to fibre therefore need to make sure interface is strong. What this means:

1. The Interface is Inherently Weak (Compared to Fibre):

- The **fibre** (e.g., carbon or Kevlar) is **very strong**.
- The **matrix** (epoxy, PEEK, etc.) is **weaker** than the fibre.
- The **interface**, where the fibre meets the matrix, is the **weakest part** of the system.
 - It's like the "glue zone" — and it's usually where things start to fail if not designed properly.

Component	Role	Examples
Fibre	Carries load, adds strength/stiffness	Glass, Kevlar, Carbon
Matrix	Binds fibres, transfers stress, protects	Epoxy, Polyester, PEEK
Interface	Enables bonding, affects durability	Modified fibre surfaces, controlled processing

What are thermosetting matrix composites, and how do their processing characteristics influence manufacturing?

Thermosetting matrix composites use a thermosetting resin (i.e., a liquid that cures irreversibly) as the binding phase to hold and transfer stress between fibres. These are the most widely used composites in structural applications.

Key Characteristics:

- Low Viscosity Resin:**
 - Thermoset resins start off as **low-viscosity liquids**.
 - This makes them **easy to flow**, allowing them to **wet the fibres completely** during layup or injection.
 - Ensures **good fibre–matrix bonding** (critical for mechanical performance).
- Then combines with Solid Fibres:**
 - Fibres (like carbon, glass, aramid) are laid into position (e.g., woven mats, roving, tapes).
 - The **liquid resin flows around the fibres**, forming the composite structure.

3. Then it undergoes curing reaction:

- The matrix then **chemically cures** (usually with heat and/or catalyst).
- This cross-linking process transforms the resin into a **rigid, thermoset plastic**.
- Once cured, the composite **cannot be reshaped** or remelted — it's **permanently set**.

4. Fibre Flexibility:

- Fibres can be pre-shaped or layered in various forms, allowing flexibility in design and tailoring of mechanical properties.

Resin Type	Key Properties
Unsaturated Polyester	Cheap, good for boat hulls, automotive
Vinyl Ester	Better chemical resistance than polyester
Epoxy	Strong, excellent adhesion, low shrinkage
Bismaleimides	High-temp resistance (aerospace)
Polyimides	Excellent thermal and chemical resistance
Phenolics	Fire resistance, often in electrical uses
Furanes & Urethane-Acrylates	Specialised roles, flexible chemistry options

Fibre forms used in thermosetting composites:

Continuous Roving:

- Consists of **thousands of continuous fibres** bundled together
- Relatively **cheap** and can be **aligned during processing**
- Used in processes like **filament winding, pultrusion**
- Provides **high directional strength** if aligned well

Chopped Glass Fibre Mats:

- **Made from short random fibres**
- **Very low-cost**
- Easy to handle and process, but produces **non-directional properties** – **AS RANDOM COMPOSITES**
- Used in **automotive panels, tubs, low-stress parts**

Continuous Filament Mats: SIMILAR TO CHOPPED

- **Continuous fibres**, but laid in **random or semi-aligned orientations**
- Slightly **stronger than chopped mats**
- More expensive than chopped fibre but **better mechanical consistency**

<i>Fibre Form</i>	<i>Key Feature</i>	<i>Benefit</i>
<i>Tri-axial Weaves</i>	<i>3-way reinforcement</i>	<i>Balanced strength, multi-axial loads</i>
<i>Non-woven Multi-axial</i>	<i>Stitched layers, no crimping</i>	<i>High performance, tailored directions</i>
<i>Braided/Knitted Preforms</i>	<i>3D shapes</i>	<i>Automated, complex geometries</i>

CONTACT MOULDING ARE HAND AND SPRAY LAY UP

Process	Method Steps
Hand Lay-Up	Place fibres manually on mould, wet with resin, roll to remove air, cure.
Spray-Up	Spray resin and chopped fibres onto mould, allow to cure.
RTM	Place dry fibres in closed mould, inject resin under pressure, cure.
Filament Winding	Pull fibres through resin bath, wind on rotating mandrel, cure.
Pultrusion	Pull fibres through resin and heated die, cure as continuous profile.
Hot Press Moulding	Place pre-filled moulding compound in hot mould, compress, cure.
Prepreg Autoclave	Lay pre-impregnated fibres in mould, consolidate in autoclave under heat and pressure.

Process	Description	Typical Products	Advantages	Disadvantages
Hand Lay-Up	Fibres manually placed on mould and wetted with resin by brush; cured	Boat hulls, storage tanks	Simple, low cost, large parts	Low speed, labour-intensive, inconsistent quality
Spray-Up	Resin and chopped fibres sprayed onto mould using spray gun	Baths, cladding panels	Fast, portable, low cost	Only random fibres, one good surface, quality varies
Resin Transfer Moulding (RTM)	Dry fibres placed in mould, resin injected under pressure	Car doors, water tanks	Good surface finish, inserts possible, suitable for volume	Lower fibre volume fraction, strength not optimal
Filament Winding	Fibres soaked in resin wound onto rotating mandrel	Tubes, fishing rods	High strength, precise fibre placement	One good surface, slow, limited shapes
Pultrusion	Fibres pulled through resin	Beams, ladders	Continuous profiles, high volume	Limited to constant cross-sections

	bath and heated die to cure			
Hot Press Moulding	Compression or injection moulding with pre-filled resin	Automotive components	High volume, short cycle time	Expensive tooling, fibre damage possible
Prepreg Autoclave	Pre-impregnated fibre sheets laid in mould, cured under heat and pressure in autoclave	Aerospace, F1 parts	Excellent quality, high performance	Very expensive, slow, high skill required

LECTURE 7

What is Casting? USES LOWER VISCOSITY LIQUIDS !!

Casting involves:

- Pouring a **liquid** (usually molten metal) into a mould cavity.
- Letting it **solidify** either:
 - By **cooling** (as with metals), or **chemical reaction** (as with thermosets/elastomers).
- Removing the solid part once it sets.

How the Mould is Filled:

1. **Gravity Casting** – Liquid flows in under its own weight (simple and cheap).
2. **Centrifugal Casting** – Spinning forces help fill the mould (for rings, pipes).
3. **Pressure Die Casting** – High pressure is used (fast and accurate for mass production).

Feature	Sand Casting	Die Casting
Mould Material	Sand	Metal (die)
Cost	Low	High (tooling)
Production Volume	One-offs or low volume	High volume (mass prod.)
Accuracy & Finish	Low	High
Material Types	Mostly metals	Mostly metals (especially aluminium, zinc)

What is Moulding? - thicker, sticky materials (unlike casting which uses low-viscosity liquids).

- The material is **hot and viscous** when pressed into the mould.
- It **solidifies on cooling** and takes the shape of the die.

Materials Used in Moulding:

- **Thermoplastics** (e.g., ABS, polypropylene) - **Thermosetting elastomers** (e.g., rubbers) - **Glasses** (e.g., lenses, light covers)

How Moulding Works:

1. The hot material is injected or pressed into the die.
2. It fills the cavity under pressure.
3. Once cooled and solidified, the mould is opened and the part is removed.

Feature	Transfer Moulding	Injection Moulding
Material Entry	Plunger pushes charge into mould	Screw feeds melted polymer into mould
Used For	Thermosets, rubbers	Thermoplastics
Suitable For	Moderate complexity	High precision, high volume
Common Products	Encapsulation, rubber gaskets	Toys, CD cases, housings, containers

What is Joining?

Joining is the process of **connecting two or more components** to form an assembly. The method used depends on the **material type**.

For Metals:

- **Bolting** – Mechanical fastening with bolts/nuts (removable)
- **Riveting** – Permanent mechanical fastening
- **Welding** – Fusing materials using heat (and sometimes pressure)
- **Brazing** – Joining with a filler metal below the base metal's melting point
- **Soldering** – Low-temperature joining (e.g. electronics)

For Polymers:

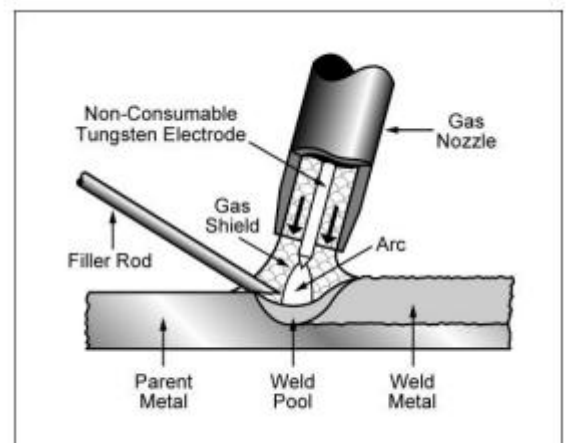
- **Snap Fasteners** – Mechanical interlocking clips
- **Thermal Bonding** – Heat melts the interface to fuse the parts
- **Ultrasonic Welding** – High-frequency vibrations generate heat at the joint

For Ceramics:

- **Diffusion Bonding** – High temp + pressure bonding without melting
- **Adhesives** – Advanced glues suitable for difficult material combinations (glass, ceramics, metals)

GAS TUNGSTEN ARC WELDING: a joining record

- It creates an arc between a **non-consumable tungsten electrode** and the workpiece.
- The weld area is protected by **inert gas** (argon, helium, or CO₂).
- **Filler material** is added separately (rod or wire).
- The tungsten electrode **does not melt** — it only generates heat.



Advantages of TIG Welding

- Extremely **clean and precise** welds.
- **Better control** than MIG, especially useful for **thin sheet metal**.
- Can be **automated**.
 - Used for **airtight/watertight** applications.
 - Low tooling cost
 - Low labour intensity

Pros	Cons
Clean, precise welds	Slower than MIG
Works with thin sheet	More expensive than MIG/MMA
Low distortion	Requires good operator skill
High-quality finish	Not ideal for thick sections

What is Induction -pic & Flame Hardening?

These are **surface-hardening processes** that:

- **Rapidly heat the outer surface** of steel into the **austenitic region**
- Then **rapidly cool it** (quench) to form **martensite**, a very hard phase

This produces:

- A **hard, wear-resistant surface**
- A **tough, machinable interior**

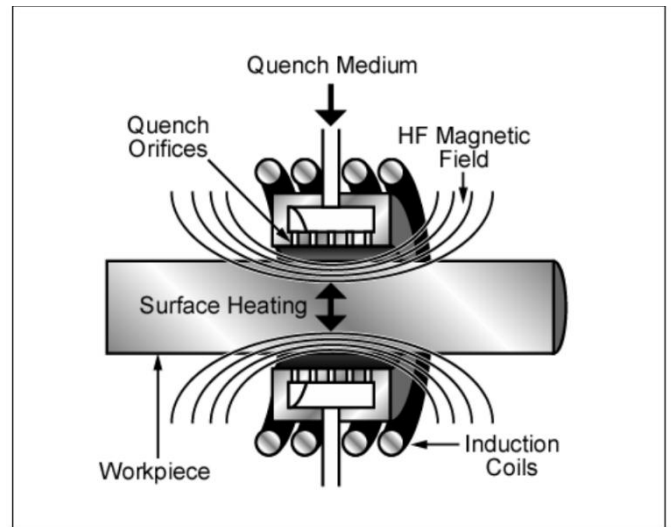
Two Techniques:

1. Induction Hardening

- Uses a **high-frequency electromagnetic field** to induce **eddy currents** in the surface
- Localised heating — depth of hardening depends on frequency
- Very **controlled, clean, and precise**

2. Flame Hardening

- Uses **gas burners** to heat the surface
- Simpler and cheaper, but less precise
- Also followed by rapid cooling



LECTURE 8

What Is a Dislocation?

- A **dislocation** is a defect in the crystal structure of a metal.
- When metals deform, dislocations **move** — so **stopping dislocations = increasing strength**.

Hardening Mechanisms for Metals

1. Solid Solution Hardening (Chemistry-Based)

What happens:

- **Solute atoms** (e.g., carbon in iron) are added to the metal.
- These atoms **distort the lattice** around dislocations, making it harder for dislocations to move.

Example:

- Adding carbon to iron to make **steel**

Key Point:

- Strengthens metal by creating local strain fields that resist dislocation motion.

2. Precipitation Hardening (Microstructure-Based)

What happens:

- **Tiny, dispersed particles (precipitates)** form inside the metal.
- Dislocations are **blocked by these particles**, forcing them to bend or break around them.

Example:

- **Aluminium-copper alloys** (like Duralumin)

Key Point:

- Excellent for heat-treatable alloys — great **strength-to-weight ratio**.

3. Work Hardening (Strain Hardening, also Microstructure-Based)

What happens:

- **Cold working** (bending, rolling, hammering) increases dislocation density.
- Dislocations **interact and tangle**, making further movement more difficult.

Step-by-Step Process

1. Solutionizing

- Heat the alloy to the **solid solution region** (above T_{ss} , below T_m)
- At this temp, **all alloying elements dissolve** in the matrix

2. Quenching

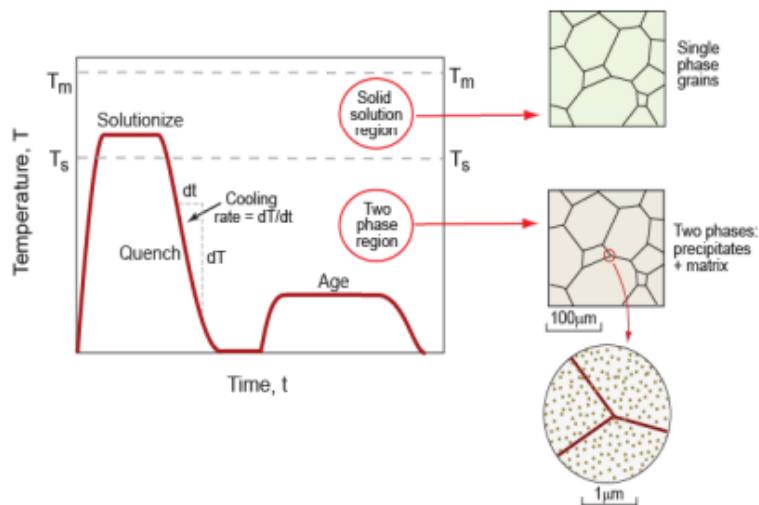
- Rapidly cool it to room temperature → freezes the structure
- Creates a **super-saturated solid solution**

3. Aging

- Reheat to a **moderate temperature**
- Over time, **tiny precipitate particles form** (the "second phase")
- These particles **impede dislocation movement**, increasing strength

Property control by **chemistry** and **microstructure**

Metals: Precipitation hardening



Examples:

*Heat-treatable Al alloys
(age hardening)*

*Carbon & alloy steels
(quench and temper)*

3. What type of bonding leads to the highest modulus in materials?

- a. Van der Waals bonding
- b. Metallic bonding
- c. Ionic or covalent bonding
- d. Hydrogen bonding

Answer: c. Ionic or covalent bonding

Why Other Options Are Wrong:

- **a. Reduces modulus** → Modulus doesn't significantly change with quenching.
- **b. Promotes grain growth** → Opposite: slow cooling promotes grain growth, not quenching.
- **d. Removes dislocations** → Quenching doesn't do this — **annealing** might.

5. What does quenching achieve in heat-treated steels?

- a. Reduces modulus
- b. Promotes grain growth
- c. Forms martensite for increased strength
- d. Removes dislocations

Answer: c. Forms martensite for increased strength

12. In precipitation hardening, what is the role of aging?

- a. To increase grain size for flexibility
- b. To dissolve precipitates into the matrix
- c. To form tiny particles that block dislocations
- d. To anneal out residual stress

Answer: c. To form tiny particles that block dislocations

Common Fillers in Polymer Composites

Filler Type	Key Properties	Effect on Composite	Used In
Talc	Plate-like, increases stiffness, reduces shrinkage, low cost	Increases modulus, reduces toughness, reduces shrinkage	Automotive parts (e.g. dashboards)
Calcium Carbonate	Inexpensive, increases stiffness, increases density	Increases modulus, increases density, low cost filler	Consumer goods, packaging, PVC pipes
Glass Beads	Spherical, improves flow, increases stiffness	Improves modulus and processability, smooth surface finish	Medical devices, electrical casings
Mica	Layered, increases dimensional stability, barrier properties	Improves stiffness, dimensional stability, and heat resistance	Paints, insulation, high-performance plastics
Carbon Black	Improves UV resistance, electrical conductivity	Increases toughness, adds conductivity and UV stability	Tyres, automotive trims, cables

Silica	Hard, abrasion-resistant, improves thermal stability	Increases hardness and wear resistance	Sealants, coatings, high-wear parts
Wood Flour	Natural, biodegradable, low cost	Reduces cost, improves stiffness, slightly reduces toughness	Decking, furniture, wood-plastic composites
Alumina Trihydrate (ATH)	Flame retardant, endothermic decomposition	Adds flame retardancy, slightly reduces toughness	Electrical housings, flame-retardant plastics

